Physiological Measurement

PAPER

Fine-tuning the ECG-FM foundation model for Chagas screening: what transferred to the hidden cohorts of the PhysioNet Challenge 2025

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Abstract

Objective. Our entry to the George B. Moody PhysioNet Challenge 2025 detected Chagas disease from the 12-lead electrocardiogram (ECG) with a frozen ECG-FM foundation model and a small head, and scored 0.245 on the hidden test cohorts. We asked how far the pretrained encoder could be adapted, and what internal evaluation on public data can show. *Approach.* We added one change at a time to the entry: soft labels, augmentation, two-stage fine-tuning with layer-wise learning-rate decay, 5 s windows matching the pretraining segments, a hard-negative ranking loss and a three-seed ensemble. Models were scored on a patient-grouped held-out split. *Main results.* Fine-tuning beat the stage-1 model it started from (TPR@5% 0.375 against 0.343); a frozen encoder trained to convergence was not clearly worse than a fine-tuned one. On the 10 s format, 5 s windows raised AUROC by $+0.043 \pm 0.010$ ($+0.077 \pm 0.023$ on CODE-15% records alone). The ranking loss removed the residual length cue (gap +0.08 to +0.01), averaging three seeds gained 0.020, and age and sex added nothing. These comparisons had to be made within recording length, because 7.34 s CODE-15% records are 12.7 times as often positive as 10.24 s ones and record length alone reaches an AUROC of 0.677. *Significance.* Scored by the organisers, the revised entry raised our hidden test mean from 0.245 to 0.289, gaining on two of three cohorts. ECG-FM can be fine-tuned for Chagas screening. Which of our changes helped only became visible once the evaluation controlled for recording length.

1 Introduction

Chagas disease, caused by *Trypanosoma cruzi*, affects millions of people in Latin America and is often silent for decades before it produces cardiomyopathy. Serological testing is the diagnostic standard but capacity is limited, so an inexpensive test that ranks people for serology is valuable. The 12-lead ECG is a natural candidate, because Chagas cardiomyopathy leaves conduction and rhythm abnormalities such as right bundle branch block and left anterior fascicular block (Jidling et al., 2023).

The George B. Moody PhysioNet Challenge 2025 asked teams to rank ECGs by the probability of Chagas disease (Reyna et al., 2025; Goldberger et al., 2000). The score is the fraction of positive cases found among the 5% of records with the highest predicted probability (TPR@5%). Training data came from three public sources with labels of very different quality: CODE-15% with self-reported labels (Ribeiro et al., 2020), SaMi-Trop with serologically confirmed positives (Cardoso et al., 2016), and PTB-XL with presumed negatives (Wagner et al., 2020). Models were scored on three hidden, serologically labelled Brazilian cohorts.

Several strong entries started from an ECG encoder pretrained on other data, and they did not agree on whether to fine-tune it. The winning team found full fine-tuning of its vision-transformer foundation model better than a head on the frozen model, with a cross-validated score of 0.490 against 0.381 (Van Santvliet et al., 2025), and the authors of ECG-FM found full fine-tuning better than a linear probe on their own tasks (McKeen et al., 2025). Another team found the opposite on hidden data, where a frozen encoder did better than a fine-tuned one (Wang et al., 2025). With

Table 1. Composition of the data by source and record length. Record length is the number of samples divided by the sampling frequency. The first group also contains the few records shorter than 5 s. Prevalence is over both splits.

		Training split		Held-out split		Prevalence
Source	Record length	Records	Positive	Records	Positive	(%)
CODE-15%	< 7.5 s	163,655	4,977	40,867	1,259	3.05
	7.5–9.9 s	5,226	6	1,273	0	0.09
	≥ 9.9 s	105,299	240	26,435	77	0.24
SaMi-Trop	< 7.5 s	564	564	147	147	100.00
	7.5–9.9 s	74	74	16	16	100.00
	≥ 9.9 s	667	667	163	163	100.00
PTB-XL	≥ 9.9 s	17,400	0	4,399	0	0.00

labels as noisy as these, fine-tuning might overwrite what pretraining taught the encoder, or it might be the only way to reach the features the disease leaves in the signal.

Our entry kept the ECG-FM encoder (McKeen et al., 2025) frozen and trained a small head on its output, with age and sex concatenated to the pooled feature (Nguyen et al., 2025a). It scored 0.245 on the hidden test set, eighth of the ranked teams. Internally, unfreezing the encoder cost us 0.08 of score, which we put down to fine-tuning overwriting what the encoder had learned. Our internal and hidden scores also disagreed: a model trained without CODE-15% scored best on our own split but 0.081 on the hidden validation cohort, against 0.323 for the entry. Other teams saw the same: the median score of the ranked teams fell from 0.279 on the hidden validation cohort to 0.236 and 0.100 on two of the test cohorts, and the best overall test score was 0.323 (Reyna et al., 2025). One team reported a variant that scored 0.20 higher locally and 0.11 lower on the hidden validation cohort (Wang et al., 2025).

Disagreement this consistent suggests that models learned a feature that predicts the label in the training data for reasons unrelated to the disease, a *shortcut* (Geirhos et al., 2020). In medical imaging, shortcuts have been traced to the hospital and the acquisition device (Zech et al., 2018; DeGrave et al., 2021). Here every SaMi-Trop record is positive and every PTB-XL record negative. The winning team warned that the original sampling frequency and the mains frequency could act as cues of this kind (Van Santvliet et al., 2025). A shortcut also stands in the way of the fine-tuning question, since a pooled internal comparison may reward whichever model exploits it best.

We went back to the entry with two questions: how much more the pretrained encoder had to give, and whether our own held-out set could tell us. We added two-stage fine-tuning with layer-wise learning-rate decay, 5 s input windows, a ranking loss and seed ensembling, one change at a time. Along the way we found that the length of a recording predicts its Chagas label inside CODE-15%, a shortcut we have not seen described, and that it inflates every pooled metric on these data. We therefore built a validation protocol that cannot reward it, selected a revised entry under that protocol, and had it scored by the organisers on the hidden cohorts (section 3.11).

2 Methods

2.1 Data and held-out split

We used the public Challenge training data: CODE-15% (400 Hz), SaMi-Trop (400 Hz) and PTB-XL (500 Hz), 366,185 records in total after conversion to the Challenge format. CODE-15% labels are self-reported, SaMi-Trop labels are serologically confirmed and all positive, and PTB-XL records are presumed negative because of their European origin (Reyna et al., 2025).

CODE-15% contains repeated examinations of the same patient, on average 1.48 per patient, and in a record-level split of our data 45.5% of test examinations shared a patient with a training examination. We therefore grouped records by patient identifier (253,720 groups) and held out 20% of the groups, stratified by source and label. The held-out set has 73,300 records and 1,662 positives. We kept its natural prevalence of 2.27%, because TPR@5% is bounded above by 0.05 divided by the prevalence and so cannot be compared between sets of different prevalence. The remaining 292,885 records form the training split. Table 1 gives the composition.

2.2 Metrics and statistics

The primary metric is the Challenge score, TPR@5%. We also report the AUROC, which is independent of prevalence. Confidence intervals are 95% percentile intervals from 1,000 bootstrap resamples of the test records (Efron and Tibshirani, 1993). To compare two configurations we used a paired bootstrap, resampling the same test records for both, and report the standard error (SE) of the difference. Unless stated otherwise, a configuration was trained with three random seeds and we report the mean and the standard deviation across seeds. A *seed ensemble* is the mean of the

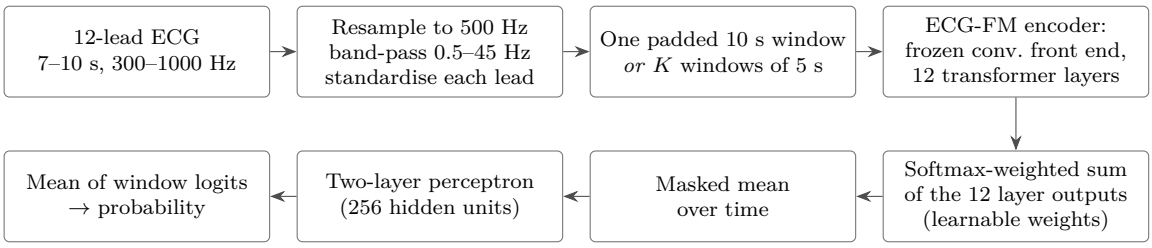


Figure 1. The model. Stage 1 trains the layer weights and the perceptron on the frozen encoder; stage 2 also fine-tunes the 12 transformer layers. The fixed-window variant averages the logits of $K = \lceil T/5\,\mathrm{s} \rceil$ evenly spaced 5 s windows of a record of length T.

predicted probabilities of the three models. The decision rules of sections 2.7 and 2.8 were fixed in writing before the runs they governed (dated entries in the repository's decision log).

2.3 The pretrained encoder and our adaptation of it

ECG-FM is a convolutional front end followed by 12 transformer layers of width 768, pretrained on 5 s segments sampled at 500 Hz with each lead standardised (McKeen et al., 2025). As in our Challenge entry, we used the released checkpoint that its authors had further fine-tuned on the diagnostic labels of MIMIC-IV-ECG (Gow et al., 2023).

The classifier wrapper released with ECG-FM detaches the encoder output from the computation graph, so we read the output of each transformer layer with a forward hook. The 12 outputs are combined with learnable softmax weights, initialised to be uniform, and averaged over time. Padded positions are excluded from the average and masked in the encoder's attention. In our Challenge entry this mask was dead code, and the zero padding of a 7.34 s record was averaged in as if it were signal. A head of dropout, a linear layer to 256 units, a ReLU, dropout and a linear layer gives one logit. None of this is in the released ECG-FM code; the hooks, the layer weights, the pooling and the head are ours.

The Challenge entry concatenated age and sex to the pooled feature. Step 4 of the ablation (section 2.5) used feature-wise linear modulation (FiLM) instead (Perez et al., 2018), as the winning team had (Van Santvliet et al., 2025): a small network, initialised to the identity, maps age and sex to a scale and a shift of the pooled feature. The final model has no demographic input.

Figure 1 shows the pipeline. Leads were put in a fixed order, resampled to 500 Hz with a polyphase filter, band-pass filtered between 0.5 and 45 Hz with a second-order, zero-phase Butterworth filter, the design used by Hong et al. (2025), notch-filtered at 50 and 60 Hz, and standardised to zero mean and unit variance over the record. The 45 Hz cut-off (step 5c) and the two notches are ours. In the baseline, the first 10 s of a record form one window, and shorter records are zero-padded at the end.

2.4 Training recipe

Training has two stages, with the schedule and learning rates of the winning entry (Van Santvliet et al., 2025). Stage 1 trains only the layer weights, the FiLM network where there is one and the head, for 2,000 optimiser steps at 2×10^{-4}, with the encoder frozen. Stage 2 unfreezes the 12 transformer layers but not the convolutional front end, as in the encoder's own fine-tuning recipe. Layer k is trained at $2 \times 10^{-5} \cdot 0.85^{12-k}$ (Howard and Ruder, 2018) and the head at 10^{-4}, with a cosine schedule and 5% warm-up. The batch is 16 with accumulation over four batches, in 16-bit mixed precision, with weight decay 0.01 and gradient clipping at 1.0. In rounds 1 and 2, 10% of the training patients (CODE-15%) or records (PTB-XL and SaMi-Trop) were held out for validation. The checkpoint with the best validation TPR@5% was kept and training stopped after three validations without improvement.

The loss is binary cross-entropy with a positive-class weight of 5, and positives were sampled five times as often as negatives. Targets are soft for the self-reported source: 0.8 and 0.2 for CODE-15% positives and negatives, 1 for SaMi-Trop and 0 for PTB-XL (Van Santvliet et al., 2025). For CODE-15% patients with several examinations the target is $0.51\,y + 0.49\,n_+/n$, where y is the record's label and n_+/n the fraction of the patient's examinations labelled positive (Hong et al., 2025). All positives were used, and negatives were subsampled so that a run saw 100,000 records in total. Most augmentations follow Hong et al. (2025) and Van Santvliet et al. (2025): resampling to and from another rate, device-like filtering, baseline wander, Gaussian noise, amplitude scaling, time warping and random masking. The mains-interference and crop-and-shift

augmentations are our own, from the Challenge code.

All training and inference ran through the unmodified Challenge entry scripts, so the code path is the one the organisers execute. Jobs ran on an NVIDIA Titan RTX with the memory allocator capped at 14.5 GiB to mimic the 16 GB card of the Challenge environment.

The front end of ECG-FM begins with a group normalisation over time, one group per channel, that ignores padding. Trailing zeros therefore shift the features at valid positions even when they are masked, as the repository's tests show for a 7 s record padded to 8 s and to 10 s. With every record padded to 10 s the distortion is a fixed function of record length that the head could learn to read, which gave us a reason to try unpadded 5 s windows that does not depend on the shortcut.

2.5 Round 1: ablation ladder

To attribute changes in performance we built the model in steps from our Challenge entry, each step adding one change to the previous one. Step 0 is the original Challenge entry: frozen encoder, age and sex concatenated to the pooled feature, hard labels, and class-balanced sampling, trained for 16 epochs. It is a reconstruction in the new code, and its layer weights are those of a frozen-encoder model trained for that long. Step 1 adds pipeline corrections (the padding mask and a vectorised band-pass filter in place of the original signal cleaner) and early stopping. Step 2 adds the soft labels and class weighting. Step 3 adds augmentation. Step 4 replaces the concatenation of age and sex with FiLM. Step 5 adds stage-2 fine-tuning. Step 5b removes age and sex altogether, and step 5c lowers the upper band-pass frequency from 100 to 45 Hz. Steps 0 to 5c were trained once, on a fixed set of 100,000 training records. The final configuration (B0) is step 5c with the negative cap applied inside the training script, as the organisers would train it, and it was trained with three seeds. For B0 we also tried stage-2 learning rates of 5×10^{-6} and 10^{-6}, the latter used by the authors of ECG-FM, with three seeds each.

2.6 Evaluation views and the record-length diagnostic

The association between record length and label (section 3.2) came to light after round 1. We then added the views below and re-scored every model on them, reading the length of each record from its WFDB header.

A All held-out records. Both the data source and the record length can be used as cues.

B CODE-15% records only, which removes the source cue.

S7 CODE-15% records shorter than 7.5 s (40,867 records, 1,259 positives). Neither cue is available.

S10 CODE-15% records of at least 9.9 s (26,435 records, 77 positives), the format of the hidden cohorts. Neither cue is available, but positives are few.

S10+ The negatives of S10 with all held-out positives of at least 9.9 s; 163 of its 240 positives are SaMi-Trop, which reintroduces a source cue. Secondary in round 2, primary in round 3.

C View A after perturbations of the kind the organisers applied to the hidden data (Reyna et al., 2025): resampling to and from 300–1000 Hz, device-like filtering, noise, baseline wander and mains interference.

The *duration-only baseline* is a predictor that outputs 1 for records shorter than 7.5 s and 0 otherwise. The *length-cue gap* is the mean predicted probability of CODE-15% negatives shorter than 7.5 s minus that of CODE-15% negatives of at least 9.9 s; it is close to zero for a model that ignores record length. Two diagnostics concern the source cue: the mean predicted probability of PTB-XL records, all of which are negative, and the *source gap*, the mean predicted probability of SaMi-Trop positives minus that of CODE-15% positives.

2.7 Round 2: input windows, matched negatives, interpolation and ensembles

Fixed 5 s windows. ECG-FM was pretrained on 5 s segments (McKeen et al., 2025). With 5 s windows, no record of at least 5 s is padded, so the network cannot observe the record length. During training one random 5 s window is drawn per record per epoch. At validation and test time $K = \lceil T/5\,\mathrm{s} \rceil$ evenly spaced windows are evaluated and their logits averaged. Random 5 s training crops with averaging over windows at inference were also used by another Challenge team, who averaged the top quarter of the window probabilities (Dong et al., 2025).

Weight interpolation. A frozen encoder outperformed a fine-tuned one on hidden data for one Challenge team (Wang et al., 2025), and fine-tuning can degrade out-of-distribution performance

(Kumar et al., 2022). We interpolated the weights of the stage-1 model (θ_1, frozen encoder) and the stage-2 model (θ_2) as $(1-\alpha)\theta_1 + \alpha\theta_2$ (Wortsman et al., 2022) and scored $\alpha = 0$, 0.5 and 1.

Length-matched negative sampling. When negatives are subsampled, they are drawn so that their distribution over three length groups (shorter than 7.5 s, 7.5 to 9.9 s, at least 9.9 s) equals that of the positives. Record length then carries no information about the label in the training or validation data. This is frequency matching of controls to cases on a confounder, the group-balancing remedy for spurious correlations (Idrissi et al., 2022; Sagawa et al., 2020), and it costs nothing here because the cap discards two thirds of the negatives anyway.

Seed ensembles. Three models that differ only in the random seed are averaged (Lakshminarayanan et al., 2017), as in earlier Chagas models and one Challenge entry (Jidling et al., 2023; Nejedly et al., 2025).

We trained three configurations derived from B0 with three seeds each: matched negatives (M1), fixed windows (W1), and both (MW, scored at the three values of α). A configuration would be adopted if its mean S7 TPR@5% exceeded that of B0 by more than the paired SE, or its S7 AUROC by more than 0.010 with S7 TPR@5% not lower by more than one SE, and in either case its S10+ AUROC did not fall by more than one SE. The length-cue gap also had to shrink for M1, W1 and MW; otherwise the change was not doing what it claimed. Views A, B and C were not decision metrics, since removing the cue should lower them.

2.8 Round 3: a validation protocol that cannot see the shortcut

Rounds 1 and 2 selected checkpoints on the pooled validation TPR@5%, which the length cue inflates, and carved the validation set after the negative cap using the training seed, so each seed validated on different records. For round 3 the validation set is drawn before the cap, at natural prevalence, from 10% of the training patients stratified by source, label and length group, with a fixed seed; PTB-XL is grouped by patient when its metadata table is available. It has 29,250 records and 681 positives and is the same for every run. Checkpoints are selected on the validation TPR@5% within 7.34 s CODE-15% records, which hold 534 of its positives against 84 of at least 9.9 s. The negative cap becomes a training prevalence of 6.5%, the value the 100,000-record cap gave on our split, so that it transfers to the organisers' larger training set.

Under this protocol we re-trained B0 and W1 with three seeds. W1 rather than MW was carried in, because the two did not differ on S10+ AUROC by more than one SE in round 2 and W1 is the simpler change with the higher S7 score, so length-matched sampling is not part of the entry. We tested three additions to W1. W1R adds the hard-negative ranking loss of Hong et al. (2025): a margin-ranking loss on the top 20% of negatives per batch by predicted probability, with margin 0.5 scaled by the difference of the soft targets and weight 2.5 relative to the cross-entropy. W1S drops PTB-XL, which exists solely as negatives and is absent from the hidden data. W1-TTA infers over 5 s windows every 2.5 s, without retraining. Two further W1 seeds tested a five-seed ensemble.

Because the hidden cohorts are 10 s recordings, the primary metric of this round is the S10+ AUROC, with the S7 TPR@5% as a guard (not lower than B0 by more than one paired SE). A variant replaces W1 if it beats W1 on S10+ AUROC by more than one SE while passing the guard, or on S7 TPR@5% by more than one SE without losing more than one SE of S10+ AUROC; ties go to the simpler configuration, and if W1 failed the guard the candidate would be B0. Overlapping windows, and five members instead of three, had to gain by more than one SE without losing elsewhere. The length-cue gap must stay near zero for every fixed-window variant, and for W1S the source gap must not grow.

3 Results

3.1 The frozen-encoder entry, re-created

Step 0 of the ladder re-creates our Challenge entry. It scored 0.412 in TPR@5% on the pooled held-out set and 0.377 within 7.34 s records (table 2). It is not the shipped model, and its scores may differ somewhat from those of the entry we submitted. It resamples with a polyphase filter rather than in the frequency domain, and it trains on 90% of the training patients, where the shipped model was trained on all of them and saved after the 16th epoch. With the encoder frozen, its head weighted the middle layers most: layer 7 received 0.129 of the weight and layer 12 only 0.040.

3.2 Record length predicts the label in CODE-15%, and every model we trained used it

Over the ablation the pooled score rose from 0.412 to 0.436, yet the steps that raised it were not the ones we had expected to matter, and looking for the reason led us to record length. CODE-15% recordings come in two lengths, 7.34 s and 10.24 s (table 1). Of the 204,522 records shorter than

Table 2. The ablation ladder, scored on the pooled views and within record-length groups. Steps 0 to 5c are single runs; B0 and the uncapped run are means over seeds.

		Pooled (A)		Pooled (B)	Within 7.34 s (S7)		10.24 s (S10)	Length-cue
	Configuration	TPR@5%	AUROC	AUROC	TPR@5%	AUROC	AUROC	gap
0	Original entry (frozen encoder)	0.412	0.847	0.851	0.377	0.807	0.720	+0.30
1	+ pipeline fixes, validation split	0.442	0.862	0.857	0.376	0.812	0.733	+0.30
2	+ soft labels, class weighting	0.427	0.856	0.850	0.365	0.802	0.711	+0.16
3	+ augmentation	0.407	0.837	0.836	0.360	0.785	0.700	+0.17
4	+ FiLM demographics	0.434	0.852	0.850	0.373	0.807	0.705	+0.14
5	+ two-stage fine-tuning	0.435	0.854	0.850	0.378	0.806	0.727	+0.13
5b	demographics removed	0.438	0.848	0.849	0.399	0.800	0.739	+0.13
5c	0.5–45 Hz band-pass	0.437	0.842	0.847	0.400	0.799	0.733	+0.12
B0	5c, negatives capped in training (3 seeds)	0.436	0.829	0.830	0.384	0.787	0.695	+0.17
	all negatives, no cap (2 seeds)	0.427	0.791	0.792	0.382	0.751	0.632	+0.07

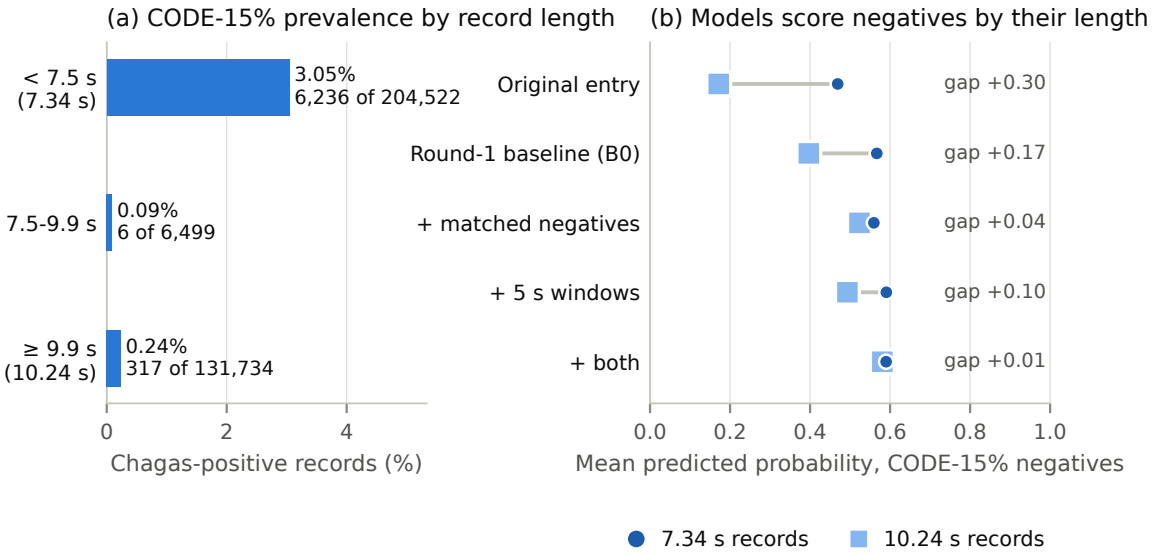


Figure 2. The length shortcut. (a) Prevalence of the Chagas label in CODE-15% by record length. (b) Mean predicted probability of held-out CODE-15% negatives of 7.34 s (circles) and 10.24 s (squares); a model that ignores length would give both the same mean. Means over seeds where there are several.

7.5 s, 3.05% are labelled positive, against 0.24% of the 131,734 records of at least 9.9 s, a ratio of 12.7. As a result, 95.1% of the CODE-15% positives are short records, and these are zero-padded in a 10 s window, where the padding may reveal their length to the network. A predictor that uses only the record length reaches an AUROC of 0.677 on view B and 0.647 on view A.

Every model, step 0 included, assigned higher probabilities to short records regardless of their label. For B0 the mean probability of held-out CODE-15% negatives was 0.57 for short records and 0.40 for long ones, a length-cue gap of +0.17 (figure 2(b)), and for step 0 the gap was +0.30. From here on we compare models within length groups.

3.3 What each adaptation did, within length groups

Within 7.34 s records the ladder is almost flat. Each step was a single run, and the bootstrap SE of TPR@5% in that group is about 0.013, so most differences between neighbouring rows are likely to be noise. The pipeline fixes of step 1 gave the highest pooled score of the ladder but left both the S7 score and the length-cue gap where they were. Soft labels and class weighting roughly halved the gap, from +0.30 to +0.16. Augmentation raised the mean predicted probability of the PTB-XL records from 0.08 to 0.32, possibly because resampling every record to and from other rates hides the 500 Hz sampling that had marked PTB-XL as a source of negatives.

Removing age and sex (step 5 to 5b) was the largest single-step change in the ladder, +0.021 in S7 TPR@5%, or about 1.6 SE. No step moved the S7 score by two standard errors. The order of the steps by pooled score is also not their order within 7.34 s records: the pipeline fixes led on view A, the steps without age and sex on S7.

3.4 Fine-tuning versus a frozen encoder

Once the cue was out of the comparison, fine-tuning beat the stage-1 model it started from. In the round-2 configuration with both remedies, the frozen encoder ($\alpha = 0$) scored 0.343 on 7.34 s

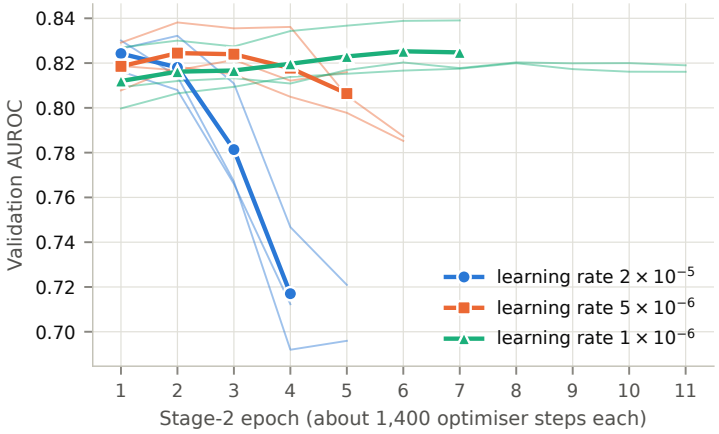


Figure 3. Validation AUROC during stage-2 fine-tuning for three learning rates. Thin lines are seeds, thick lines their mean over the epochs all three completed.

Table 3. Effect of the stage-2 learning rate on configuration B0 (three seeds, mean ± standard deviation). The ensemble column is the score of the average of the three models. Epochs are counted from one.

Stage-2	Pooled (A)		Within 7.34 s (S7)			Best stage-2
learning rate	TPR@5%	AUROC	TPR@5%	AUROC	Ensemble TPR@5%	epoch
2×10^{-5}	0.436 ± 0.010	0.829	0.384 ± 0.017	0.787	0.404	1.7
5×10^{-6}	0.423 ± 0.002	0.834	0.381 ± 0.006	0.792	0.396	2.7
1×10^{-6}	0.404 ± 0.004	0.830	0.362 ± 0.008	0.784	0.375	6.7

records and the fine-tuned one ($\alpha = 1$) 0.375, and every seed moved the same way. The AUROC rose from 0.775 to 0.795 within 7.34 s records and from 0.797 to 0.841 on the 10 s format. The halfway interpolation ($\alpha = 0.5$) scored 0.370 and offered nothing over full fine-tuning.

The frozen arm of this comparison is the stage-1 model of the same run, a head trained for 2,000 steps, not a frozen-encoder model trained to convergence. Step 0 is such a model, and within 7.34 s records it scored 0.377, close to the fine-tuned B0 at 0.384. What we can show is that stage 2 improves on stage 1, not that a fine-tuned ECG-FM beats every frozen one. The direction agrees with the winner's cross-validation and with the comparison of a linear probe and full fine-tuning by the authors of ECG-FM (Van Santvliet et al., 2025; McKeen et al., 2025). It reverses what we concluded during the Challenge from a pooled comparison with single-stage fine-tuning (Nguyen et al., 2025a), and the two-stage schedule may explain the difference, though we did not re-run single-stage fine-tuning to check.

3.5 Demographics: concatenation, FiLM, none

Replacing concatenation with FiLM, on top of augmentation, raised the pooled score, and on its own it already lowered the mean probability of PTB-XL records from 0.32 to 0.24; within 7.34 s records the change was about one standard error. Fine-tuning the encoder while it received age and sex then drove the PTB-XL mean from 0.24 to 0.01. Every PTB-XL record is negative and comes from a European cohort with its own mix of age and sex, so the model appears to have learned where the records came from. Removing age and sex returned the value to 0.07 with no loss within length groups, and the final model has no demographic input.

3.6 Learning rate, training-set size and stage-2 dynamics

At the base learning rate the validation score peaked at the first or second stage-2 epoch in every run and then declined (figure 3). Lowering the stage-2 learning rate moved the peak later, from epoch 1.7 on average to 6.7 at 10^{-6}, but did not raise it, and the test score fell with the learning rate (table 3). We had expected the opposite and kept 2×10^{-5}. The rate used by the authors of ECG-FM, 10^{-6}, was the worst of the three: 0.362 within 7.34 s records, against 0.384 at the base rate.

Training on all negatives instead of the capped 100,000 records lowered the AUROC within short records from 0.787 to 0.751, while TPR@5% on view A changed little. Validating every 500 steps, so as not to miss the early peak on the larger set, did not recover it, so the loss seems to come from the larger set itself rather than from a missed peak, and every later configuration kept the cap.

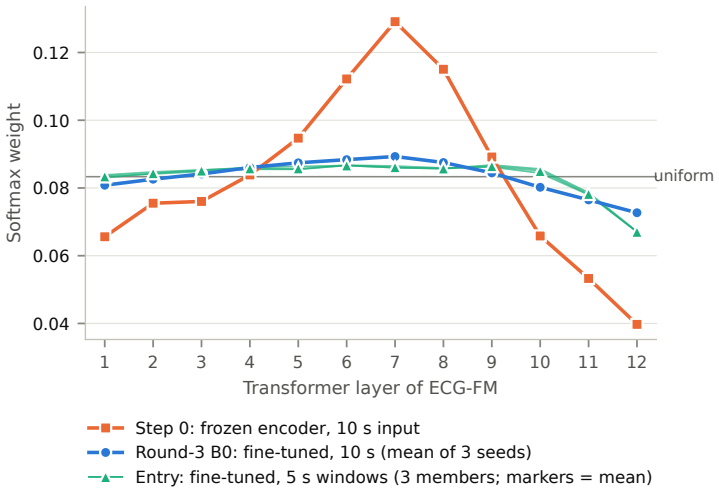


Figure 4. Learned softmax weights over the 12 transformer layers of ECG-FM. Step 0 uses a frozen encoder; round-3 B0 (mean of three seeds) and the entry's three members are fine-tuned. The horizontal line is the uniform weight, 1/12.

3.7 Which layers the head uses

With ECG-FM frozen, the head put most of its weight on layers 5 to 9 and least on the last layer (figure 4). In step 0, layer 7 carried 3.2 times the weight of layer 12. Nguyen, Phan and colleagues (2025b), working on a different ECG transformer, found middle layers better than the last, and the checkpoint we used had been fine-tuned end to end on the MIMIC-IV-ECG labels, which tends to bend the last layer towards those labels.

Once the transformer layers were fine-tuned, the mixture stayed close to uniform. Across all 42 two-stage checkpoints the largest weight lay between 0.086 and 0.095 and the smallest, always on layer 12, between 0.066 and 0.076. The shape was the same for 10 s inputs and 5 s windows, and the three entry members differed by at most 0.001 on any layer. This does not show that a fine-tuned model uses all layers alike. The weights start from uniform and received far fewer optimiser steps than in step 0, and the runs that spent longest in stage 2 moved furthest from uniform, so the flat profile may be an artefact of a short stage 2.

3.8 Fixed 5 s windows and matched negatives

Matched negatives shrank the length-cue gap from +0.17 to +0.04, and the two remedies together to +0.01. Fixed windows alone took it only to +0.10, so some of the cue survived even when the network could not see the record length. The two formats may come from different recording devices, which could leave their own marks on the signal.

The pooled score fell when the cue went: TPR@5% on view A dropped from 0.436 to 0.401 with matched negatives and to 0.403 with both. Within 7.34 s records no remedy raised TPR@5%, and the largest change was a fall of -0.009 ± 0.005 with both (table 4). Fixed windows raised the AUROC there by +0.008, a small but consistent gain. No configuration met the adoption rule on the primary view.

The 10 s format is where the windows paid off. Fixed windows raised the S10+ AUROC by $+0.056 \pm 0.010$, to 0.835. Matched negatives, which leave the input unchanged, gained less ($+0.037 \pm 0.009$), and the two together gained $+0.062 \pm 0.011$. On S10 alone the AUROC rose from 0.695 to 0.773 with both, while the Challenge score on S10+ did not change significantly.

The remedies may also have made the seeds agree: the standard deviation of the S7 score across seeds was 0.017 for B0 and 0.004 with fixed windows, although three seeds give a poor estimate of spread. Fixed windows also cut peak stage-2 memory from 4.10 to 2.81 GiB.

3.9 Seed ensembles and perturbations

Averaging three seeds raised the baseline's TPR@5% within short records from 0.384 to 0.404, and its AUROC there from 0.787 to 0.801. On S10+ the ensembles of fixed windows and of the combined configuration reached AUROCs of 0.857 and 0.854, against 0.801 for the baseline ensemble. Within 7.34 s records, nothing we changed in the model gained as much as averaging three seeds.

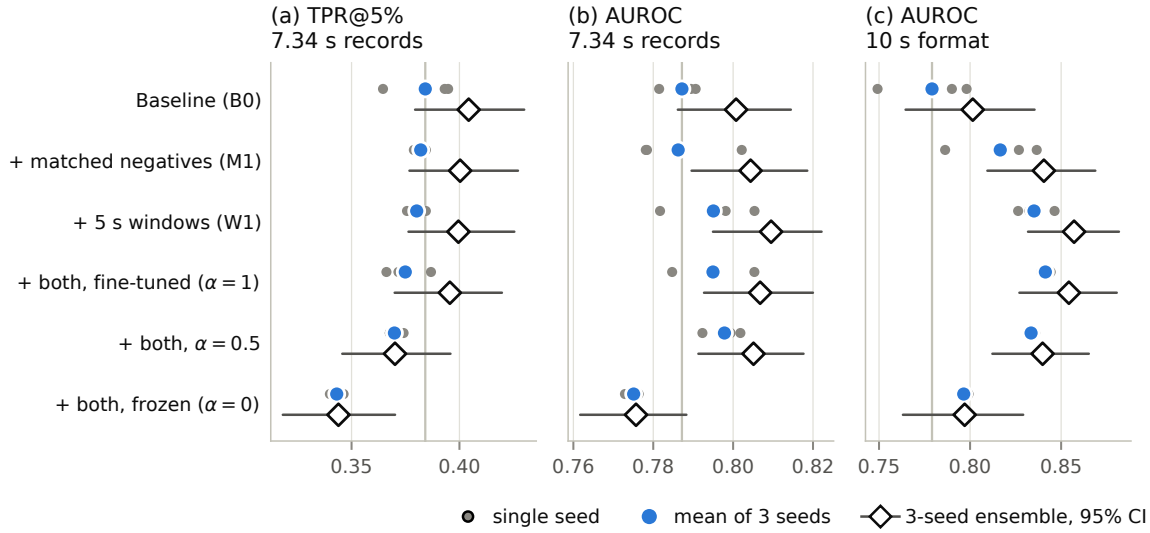


Figure 5. Removing the length cue (round 2; three seeds per row). The vertical line marks the baseline mean. (a) and (b) use held-out CODE-15% records of 7.34 s; (c) uses the 10 s format (S10+).

Table 4. Round 2 (three seeds each). Differences from the baseline mean, with the SE of a paired bootstrap over test records. The pooled score is TPR@5% on view A.

	Pooled	Within 7.34 s records (S7)				10 s format (S10+)		Length-
Configuration	TPR@5%	TPR@5%	Δ	AUROC	Δ	AUROC	Δ	cue gap
Baseline (B0)	0.436	0.384	–	0.787	–	0.779	–	+0.17
+ matched negatives (M1)	0.401	0.382	-0.002 ± 0.005	0.786	-0.001 ± 0.003	0.817	$+0.037 \pm 0.009$	+0.04
+ 5 s windows (W1)	0.428	0.380	-0.004 ± 0.004	0.795	$+0.008 \pm 0.002$	0.835	$+0.056 \pm 0.010$	+0.10
+ both, fine-tuned ($\alpha = 1$)	0.403	0.375	-0.009 ± 0.005	0.795	$+0.008 \pm 0.003$	0.841	$+0.062 \pm 0.011$	+0.01
+ both, $\alpha = 0.5$	0.383	0.370	-0.014 ± 0.007	0.798	$+0.011 \pm 0.004$	0.833	$+0.054 \pm 0.012$	+0.01
+ both, frozen ($\alpha = 0$)	0.354	0.343	-0.041 ± 0.009	0.775	-0.012 ± 0.006	0.797	$+0.017 \pm 0.017$	+0.00

The perturbations of view C appear to have had little effect once the comparison was made within length groups: TPR@5% within short records was 0.382 for B0 and 0.376 for the combined configuration on the perturbed records, close to the unperturbed values.

3.10 Round 3: the candidate under the fixed validation protocol

Under the fixed protocol the baseline scored 0.385 within short records and 0.781 in AUROC on the 10 s format, close to its round-2 values, so the protocol by itself changed little (table 5). The fixed-window model again left the S7 score unchanged (-0.002 ± 0.005) and raised the S10+ AUROC by $+0.043 \pm 0.010$. This time it met the rule, whose primary view was now the 10 s format.

All three additions passed against the baseline, but none beat W1 on the primary metric by more than one SE (figure 6). On the S7 score the ranking loss gained $+0.006 \pm 0.005$ over W1 without losing on S10+, so it qualified by the second clause of the rule. W1S did not, and overlapping windows cannot change S7, because a 7.34 s record gives the same two windows either way. Applied as written, the rule made W1R the entry, on a margin of 1.3 standard errors that would not convince us by itself; what persuaded us was the gap.

The ranking loss cut the length-cue gap from +0.08 to +0.01, although the network could not see the record length either way. Its three-seed ensemble reached 0.410 on S7, the highest S7 score of any ensemble in the comparisons of rounds 1 to 3; on S10+ its 0.841 is in line with, not above, the round-2 fixed-window ensembles (0.857 and 0.854). On S10 alone, which has no SaMi-Trop records, the AUROC rose from 0.675 for the baseline to 0.759 for W1R. Dropping PTB-XL did not make the positives more separable by source (table 5). The five-seed W1 ensemble gained $+0.003 \pm 0.003$ on S10+ over three seeds, inside one SE, which gave us no reason to train more members.

The post-Challenge entry is the three members of W1R trained inside a single call of the Challenge training script. It scored 0.416 on S7 (AUROC 0.808) and 0.854 in AUROC on S10+, within the confidence interval of the round-3 W1R ensemble. One difference from the round-3 runs is deliberate. The PTB-XL metadata table that groups records by patient is not part of the Challenge data, so the entry, run as the organisers run it, groups PTB-XL records individually, with a validation set of 29,273 records instead of 29,250; this changes which negatives are held out but no label. The three members trained in 4.1 h on a Titan RTX with a 14.5 GiB memory cap, a

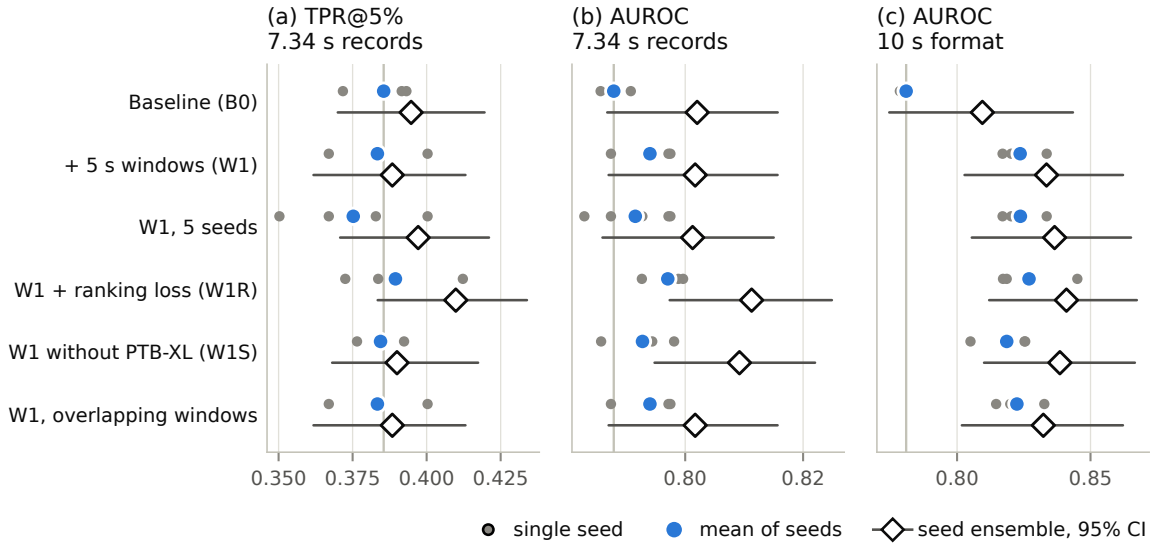


Figure 6. Round 3 under the fixed validation protocol, laid out as figure 5; B0 and W1 are the round-3 runs, and one row has five seeds.

Table 5. Round 3 (fixed validation protocol; three seeds unless stated). B0 and W1 are re-trained and differ from the round-2 runs of the same name. Differences are from the baseline mean, with the SE of a paired bootstrap over test records.

	Pooled	Within 7.34 s records (S7)				10 s format (S10+) AUROC			Length-	Source
Configuration	TPR@5%	TPR@5%	Δ	AUROC	Δ	mean	Δ	ensemble	cue gap	gap
Baseline (B0)	0.427	0.385	–	0.788	–	0.781	–	0.809	+0.15	−0.04
+ 5 s windows (W1)	0.422	0.383	−0.002 ± 0.005	0.794	+0.006 ± 0.002	0.824	+0.043 ± 0.010	0.834	+0.08	−0.02
W1 with 5 seeds	0.417	0.375	−0.010 ± 0.005	0.792	+0.004 ± 0.002	0.824	+0.043 ± 0.010	0.837	+0.08	−0.02
W1 + ranking loss (W1R)	0.425	0.389	+0.004 ± 0.005	0.797	+0.009 ± 0.003	0.827	+0.046 ± 0.012	0.841	+0.01	−0.01
W1 without PTB-XL (W1S)	0.414	0.384	−0.001 ± 0.005	0.793	+0.005 ± 0.003	0.819	+0.038 ± 0.011	0.839	+0.07	−0.01
W1, windows every 2.5 s	0.422	0.383	−0.002 ± 0.005	0.794	+0.006 ± 0.002	0.822	+0.041 ± 0.010	0.832	+0.09	−0.02

projected 10 h on the organisers' T4 if it is 2.5 times slower, against the 72 h limit. Inference over 2 × 73,300 records took 0.9 h.

3.11 External scores on the hidden cohorts

The organisers ran the training and inference scripts of the post-Challenge entry (commit 355ff92, submission F28 of the focus issue) and scored it on the hidden cohorts on 19 October 2026. Table 6 compares it with our original entry, scored on the same cohorts in 2025.

On the test mean the post-Challenge entry scored 0.289 against 0.245, a difference of +0.044. The gain came from two cohorts: REDS-II rose from 0.329 to 0.371 and SaMi-Trop 3 from 0.281 to 0.371. ELSA-Brasil did not move: both entries scored 0.125 there, as did the winning entry (PhysioNet, 2025), and TPR@5% appears to take only a few values on that cohort. For context, the highest official test mean was 0.323 (REDS-II 0.468, SaMi-Trop 3 0.376, ELSA-Brasil 0.125), the second 0.283 and the median 0.217 (Reyna et al., 2025; PhysioNet, 2025). A score of 0.289 would lie between the first and the second, but an entry submitted after the Challenge closed is not ranked, and it was built with the published methods of other teams.

Every hidden AUROC was below the 0.854 the entry reached on the 10 s format of our held-out set, the lowest being 0.582 on ELSA-Brasil; an earlier network reached 0.68 and 0.59 on its own versions of REDS-II and ELSA-Brasil (Jidling et al., 2023). The direction of the gain matches the within-format internal comparisons. Within 7.34 s records the entry, a three-model ensemble,

Table 6. Scores on the hidden data, as returned by the organisers, for our 2025 entry (Nguyen et al., 2025a) and the post-Challenge entry (W1R, three seeds). The test mean, over the three test cohorts, ranks entries in the Challenge. No confidence intervals are returned, and AUROC only for the post-Challenge entry.

	Original entry (2025)	Post-Challenge entry		Difference
Cohort	TPR@5%	TPR@5%	AUROC	in TPR@5%
Hidden validation	0.323	0.477	0.794	+0.154
REDS-II (test)	0.329	0.371	0.745	+0.042
SaMi-Trop 3 (test)	0.281	0.371	0.780	+0.090
ELSA-Brasil (test)	0.125	0.125	0.582	0.000
Test mean	0.245	0.289	–	+0.044

found more positives in its top 5% than the single step-0 model (0.416 against 0.377) at the same AUROC (0.808 against 0.807); within the 10 s format it ranked better (S10 AUROC 0.765 against 0.720). The lower absolute level was to be expected, since the held-out set was our selection set and the hidden cohorts are a different population with serological labels and their own devices.

4 Discussion

4.1 Fine-tuning ECG-FM

During the Challenge we kept ECG-FM frozen because unfreezing it lowered our internal score. With the length cue out of the comparison and a two-stage schedule, fine-tuning beat the stage-1 model it started from, though not clearly a frozen encoder trained to convergence; that comparison, with several seeds and the cue controlled, remains to be made. Whatever stage 2 contributed, it contributed early. At the base rate the validation score peaked at epoch 1.7 on average and then declined, which suggests that the encoder needs only a small push from its pretrained state. We were wrong about the learning rate and about the data: a lower stage-2 rate did not help, and training on every negative was worse than training on a third of them. That interpolating towards the frozen weights did not help was a smaller surprise. For anyone fine-tuning ECG-FM on a task like this one, our results point to a short stage 2 at a base rate near 2×10^{-5} rather than the 10^{-6} its authors used, validated often enough to catch a peak in the first or second epoch, on a capped set of negatives; why the cap helps we cannot say, since validating more often on the full set did not recover the loss.

The input and the head told us less about the encoder than we had hoped. The input that worked best had the segment length ECG-FM was pretrained on, and 5 s windows also avoid the padding that the group normalisation turns into a function of record length and hide the length from the network. Our design cannot separate these three effects, and matched negatives, which left the input as it was, also gained on the 10 s format, so part of the gain is likely to be the removal of the cue. On a frozen encoder the head favoured the middle layers; fine-tuned, its weights stayed near uniform, which may reflect no more than the few steps they received. Reading all 12 layers adds twelve scalar weights and no measurable compute, so we would keep it, though we have not shown that it beats the last layer. Age and sex we would leave out. A demographic side input can encode the data source, since age and sex differ between the European PTB-XL cohort and the Brazilian ones, and FiLM with fine-tuning made PTB-XL records easier to recognise without making Chagas easier to find. The winning team used FiLM (Van Santvliet et al., 2025), and our null result need not contradict theirs: in our ladder FiLM raised the pooled score at step 4, on a frozen encoder, but once the encoder was fine-tuned, removing age and sex cost nothing, so stage 2 seems to have learned whatever FiLM had supplied. With a frozen or differently trained encoder, demographics might still help.

The ranking loss did the one thing the windows could not. With fixed windows the network cannot see the record length, yet a gap of +0.08 remained, and the hard-negative ranking loss of Hong et al. (2025), a change to the loss rather than to the data, brought it to +0.01 for a reason we cannot give. Our guess is that the negatives it penalises most are the confidently scored short records. Under the cue, the negatives with the highest predicted probabilities in a batch are disproportionately 7.34 s records, so a loss that pushes those negatives below the positives could be removing what is left of the format signal. We did not tabulate the record lengths of the negatives the loss selected, and the explanation remains a hypothesis.

4.2 A shortcut inside the largest training source

The length of a CODE-15% recording is a strong predictor of its Chagas label: used alone, it reaches an AUROC of 0.677 among CODE-15% records. We do not know the cause. The two lengths probably correspond to two recording systems or software versions in the telehealth network that collected the data (Ribeiro et al., 2020), deployed in regions or periods with different prevalence of Chagas disease. That could also explain why a model that cannot see the record length still partly separates the formats. The public data carry no device or site information, so we could not test this.

The cue misled our own model development. Scoring within one length group lowered the AUROC of every ladder step by a similar amount (figure 7); for B0 it was 0.830 on view B, 0.787 within short records and 0.695 within long ones, the last close to the 0.68 and 0.59 an earlier model obtained on REDS-II and ELSA-Brasil (Jidling et al., 2023), although with only 77 positives its interval is too wide to rank models. B0, the end of the ladder, scored 0.023 higher on view A than step 0, and four of the eight steps raised that score. Within short records the two scored 0.377 and 0.384 ± 0.017, well inside the bootstrap SE, and B0's AUROC there was lower (0.787 against

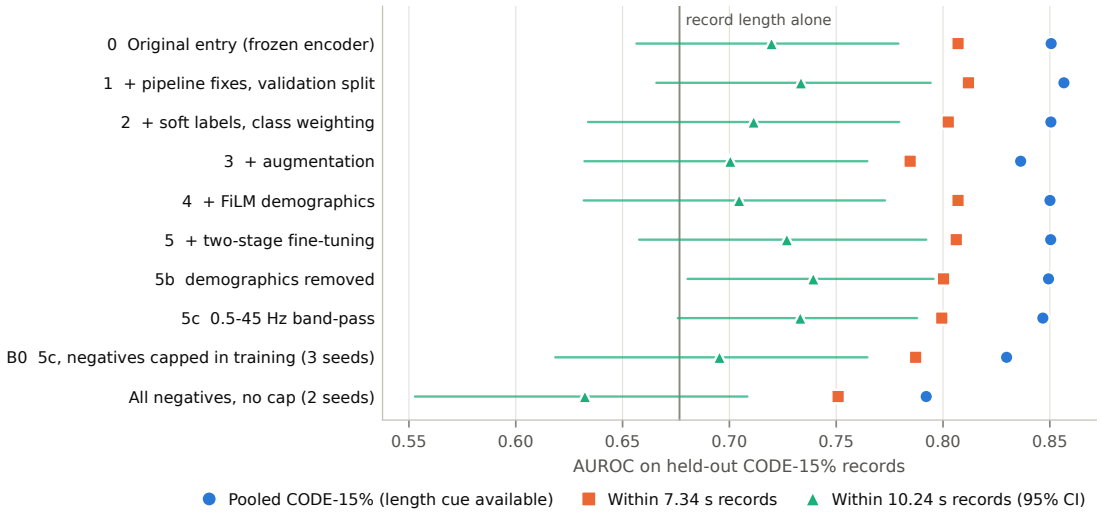


Figure 7. AUROC of each ablation step on held-out CODE-15% records: pooled (view B), within 7.34 s records (S7) and within 10.24 s records (S10, with 95% bootstrap intervals). The vertical line is the AUROC of record length alone on view B.

0.807). The pooled gain does not appear to have been a gain in detecting Chagas disease from ECG morphology, and a patient-grouped split at natural prevalence, both of which we consider necessary, did not protect against it.

Whatever its origin, the association cannot help on the hidden cohorts, whose recordings are all about 10 s long (Reyna et al., 2025). It may contribute to the gap between internal and hidden scores that almost every team observed, but probably only a few hundredths of it: the length cue raises step 0's AUROC on CODE-15% records by about 0.044 (view B against S7). The rest likely has other causes, such as cohort, label quality and device. Two of the three highest-ranked entries used augmentations that blur the cue. The winning team cropped records at random to between 5.65 and 10 s and zero-padded them (Van Santvliet et al., 2025), and the third-placed team masked the beginning and end of records "to simulate shorter signals" (Hong et al., 2025). Both make the amount of padding less informative, although neither team described the association with the label. In our Challenge paper we listed random padding or windowing as future work (Nguyen et al., 2025a).

4.3 One external score

We have one scored entry to go on. It shows that the revised pipeline, trained by the organisers with the unmodified entry scripts, found more positives among its top 5% than our original entry on two of the three hidden test cohorts, in the direction the internal within-format evidence predicted. It cannot show which change is responsible. The entry differs from the original in many ways at once, from the pipeline corrections, soft labels, augmentation and fine-tuning of the ablation to the fixed windows, the ranking loss, the validation protocol and the seed ensemble, and the hidden score tests them only together. We cannot attribute the external gain to any one of them, fine-tuning included, nor say whether dropping age and sex mattered on the hidden data. Without confidence intervals or access to the hidden data we also cannot say how much of the difference could be sampling variation; for scale, the bootstrap SE of TPR@5% on our 7.34 s held-out records was about 0.013. A second scored entry that differs from this one in a single change, for example the round-3 baseline, would be needed to attribute the gain.

4.4 Recommendations

For work on these data we suggest the following.

1. Treat the recording format as a confounder from the start. For CODE-15%, score 7.34 s and 10.24 s records separately, use the duration-only baseline and the length-cue gap as cheap checks, and expect pooled scores to fall when the cue is removed. Length-matched negatives cost nothing when negatives are abundant, and a fixed window that never needs padding hides the length, but neither guarantees that the cue is gone.

2. Give a pretrained encoder its pretraining format, here 5 s at 500 Hz, and check how its front end treats padding.

3. When fine-tuning ECG-FM, keep the second stage short and validate it often; in our runs the useful part was over within two epochs.

4. Average several seeds. It was the most reliable improvement we found, and it reduces the noise that dominates comparisons of single runs.

4.5 Limitations

Only one revised entry was scored on the hidden cohorts. The 10.24 s group of our held-out set has only 77 CODE-15% positives, and most positives of S10+ are SaMi-Trop records, which may be recognisable by their source, although the round-3 gain also appeared on S10 alone. All results come from one held-out split and one backbone, and we trained no model without a foundation-model encoder for comparison. The ablation steps before B0 were single runs. CODE-15% labels are self-reported, which limits what any internal evaluation can show.

The attribution of the length association to recording devices is a hypothesis, and so is our account of the ranking loss. The frozen arm of the fine-tuning comparison is the stage-1 model of the same run rather than a frozen encoder trained to convergence. We never trained a last-layer-only head, so the layer aggregation was not tested against the simpler choice, and the near-uniform layer weights of the fine-tuned models are confounded with the number of optimiser steps those weights received. The round-3 rule was fixed before the round-3 runs but after round 2 had shown the 10 s-format gain. Length-matched sampling was not re-tested under the fixed protocol. The held-out split was the selection set for every decision across the three rounds, so the entry's held-out scores are optimistic and are not an estimate of external performance; that is why the entry went to the organisers for scoring.

5 Conclusion

Our Challenge entry kept ECG-FM frozen. We fine-tuned its transformer layers in two stages, with a learning rate that decays towards the input, fed it 5 s windows of the length it was pretrained on, added a hard-negative ranking loss and averaged three seeds. Within 7.34 s records of our held-out data the result, a three-model ensemble, found more positives in its top 5% than the single step-0 model (0.416 against 0.377) at the same AUROC; within the 10 s format it ranked positives better (S10 AUROC 0.765 against 0.720). Those comparisons became possible only once we had found that record length predicts the Chagas label in CODE-15%, a cue that pooled scores reward and that scoring within recording formats takes out of the evaluation. Scored by the organisers on the hidden test cohorts, the revised entry raised our Challenge score from 0.245 to 0.289, with gains on REDS-II and SaMi-Trop 3 and none on ELSA-Brasil.

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To be completed by the authors.

Author contributions

To be completed by the authors.

Data availability

The CODE-15%, SaMi-Trop and PTB-XL data are publicly available through the Challenge (Reyna et al., 2025). The code, the configuration of every run, the record-length table and the scripts that generate each figure, table and number in this paper are in the project repository (to be made public by the authors); the organisers' scores of both entries are in `EXTERNAL_SCORES.csv`, and the learned layer weights of every checkpoint are in `experiments/results/entry_final_r3/layer_weights.csv`. The decision rule of each round and the reason for every change of plan are recorded, with dates, in the repository's `DECISIONS.md`, `PLAN.md`, `PLAN_R2.md` and `PLAN_R3.md`.

References

Clareci Silva Cardoso, Ester Cerdeira Sabino, Claudia Di Lorenzo Oliveira, Lea Campos de Oliveira, Ariela Mota Ferreira, Edecio Cunha-Neto, Ana Luiza Bierrenbach, João Eduardo Ferreira, Desirée Sant'Ana Haikal, Arthur L. Reingold, and Antonio Luiz P. Ribeiro. Longitudinal study of patients with chronic Chagas cardiomyopathy in Brazil (SaMi-Trop project): a cohort profile. *BMJ Open*, 6:e011181, 2016.

Alex J. DeGrave, Joseph D. Janizek, and Su-In Lee. AI for radiographic COVID-19 detection selects shortcuts over signal. *Nature Machine Intelligence*, 3:610–619, 2021.

Tianzheng Dong, Xinqi Bao, Jia Bi, and Saikat Chatterjee. Domain-adversarial pretrained encoder for ECG-based Chagas disease screening. In *Computing in Cardiology*, volume 52, 2025. Team FjordNet. doi:10.22489/CinC.2025.091.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, New York, 1993.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.

Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation*, 101:e215–e220, 2000.

Brian Gow, Tom Pollard, Larry A. Nathanson, Alistair Johnson, Benjamin Moody, Chrystinne Fernandes, Nathaniel Greenbaum, Jonathan W. Waks, Parastou Eslami, Tanner Carbonati, Ashish Chaudhari, Elizabeth Herbst, Dana Moukheiber, Seth Berkowitz, Roger Mark, and Steven Horng. MIMIC-IV-ECG: diagnostic electrocardiogram matched subset (version 1.0). PhysioNet, 2023. doi:10.13026/4nqg-sb35.

Casey Hong, Chun-Ti Chou, and Sergio González. Managing label uncertainty in the detection of Chagas disease from the ECG. In *Computing in Cardiology*, volume 52, 2025. doi:10.22489/CinC.2025.100.

Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. In *Proc. 56th Annual Meeting of the Association for Computational Linguistics*, pages 328–339, 2018.

Badr Youbi Idrissi, Martin Arjovsky, Mohammad Pezeshki, and David Lopez-Paz. Simple data balancing achieves competitive worst-group-accuracy. In *Proc. Conf. on Causal Learning and Reasoning*, volume 177 of *PMLR*, pages 336–351, 2022.

Carl Jidling, Daniel Gedon, Thomas B. Schön, Claudia Di Lorenzo Oliveira, Clareci Silva Cardoso, Ariela Mota Ferreira, Luana Giatti, Sandhi Maria Barreto, Ester C. Sabino, Antonio L. P. Ribeiro, and Antônio H. Ribeiro. Screening for Chagas disease from the electrocardiogram using a deep neural network. *PLoS Neglected Tropical Diseases*, 17:e0011118, 2023.

Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *Int. Conf. on Learning Representations*, 2022.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Kaden McKeen, Sameer Masood, Augustin Toma, Barry Rubin, and Bo Wang. ECG-FM: an open electrocardiogram foundation model. *JAMIA Open*, 8:ooaf122, 2025.

Petr Nejedly, Martina Dankova, Radovan Smisek, Ivo Viscor, Pavel Jurak, and Filip Plesinger. Knowledge distillation from general multi-objective ECG classification model for Chagas disease detection in 12-lead ECGs. In *Computing in Cardiology*, volume 52, 2025. doi:10.22489/CinC.2025.132.

Kelvin Nguyen, Andy Smithwick, Maxwell Loetscher, Shadi Manafi, Zaniar Adarlan, and Saman Parvaneh. Finetuning foundational ECG models to detect Chagas disease. In *Computing in Cardiology*, volume 52, 2025a. doi:10.22489/CinC.2025.416.

Phu X. Nguyen, Huy Phan, Hieu Pham, Christos Chatzichristos, Bert Vandenberk, and Maarten De Vos. Exploiting a mixture-of-layers in an electrocardiography foundation model. arXiv:2509.00102, 2025b. Title of version 1; later versions are titled "ECG-Soup: harnessing multi-layer synergy for ECG foundation models".

Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: visual reasoning with a general conditioning layer. In *Proc. AAAI Conf. on Artificial Intelligence*, volume 32, 2018.

PhysioNet. George B. Moody PhysioNet Challenge 2025: results, 2025. URL https://moody-challenge.physionet.org/2025/results/. Accessed 2026-10-30.

Matthew A. Reyna, Zuzana Koscova, Jan Pavlus, James Weigle, Soheil Saghafi, Paulo Gomes, Andoni Elola, Mohammad Sina Hassannia, Katherine Campbell, Ali Bahrami Rad, Antônio H. Ribeiro, Antonio L. P. Ribeiro, Reza Sameni, and Gari D. Clifford. Detection of Chagas disease from the ECG: the George B. Moody PhysioNet Challenge 2025. In *Computing in Cardiology*, volume 52, pages 1–4, 2025. arXiv:2510.02202.

Antônio H. Ribeiro, Manoel Horta Ribeiro, Gabriela M. M. Paixão, Derick M. Oliveira, Paulo R. Gomes, Jéssica A. Canazart, Milton P. S. Ferreira, Carl R. Andersson, Peter W. Macfarlane, Wagner Meira, Thomas B. Schön, and Antonio Luiz P. Ribeiro. Automatic diagnosis of the 12-lead ECG using a deep neural network. *Nature Communications*, 11:1760, 2020.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: on the importance of regularization for worst-case generalization. In *Int. Conf. on Learning Representations*, 2020.

Lore Van Santvliet, Phu Xuan Nguyen, Bert Vandenberk, and Maarten De Vos. Detecting Chagas disease using a vision transformer-based ECG foundation model. In *Computing in Cardiology*, volume 52, 2025. doi:10.22489/CinC.2025.019.

Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Fatima I. Lunze, Wojciech Samek, and Tobias Schaeffter. PTB-XL, a large publicly available electrocardiography dataset. *Scientific Data*, 7:154, 2020.

X. Wang, A. Syversen, Z. Ding, et al. Two-stage domain adversarial learning to identify Chagas disease from ECG and patient demographic data. In *Computing in Cardiology*, volume 52, 2025. doi:10.22489/CinC.2025.158.

Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo-Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. In *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, pages 7959–7971, 2022.

John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. *PLoS Medicine*, 15:e1002683, 2018.